Stress Induced Failure Mechanism in NAND Flash Memory Devices

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Abstract:

NAND Flash memories have emerged as the technology of choice for non-volatile memory storage due to their high storage density, low cost and small form factor. Typical Flash memory cells, as illustrated in Fig. 1(a) however have a limited endurance, typically of the order of 10^5 cycles for 2D SLC NAND devices. Programming and Erasing NAND cells required tunneling of electrons across the tunnel oxide layer into the floating gate oxide layer. When performed repeatedly, this process introduces defects in the oxide layer due to trap generation as illustrated in Fig. 1 (b). By default, cells are present in the '1' state, i.e., no charge is stored in the floating gate. Therefore, programming a cell from '1' to '0' induces a higher stress on the oxide layer compared to programming to '1'. Therefore, if data is repeatedly programmed and erased, the cells being programmed to '0' encounter more damage to the oxide layer and consequently, the analog properties of the cell, most notably the threshold voltage V_{th}, are irreversibly changed. Fig. 2 illustrates the process of generation of bad cells with oxide damage. Here we investigate the effects of repeated Program-Erase (PE) cycles and Program Stress (PS) cycles on the Fail Bit Count (FBC) of NAND Devices.

Measurement Setup:

The experiments were conducted on a Micron 2D SLC NAND (29F4G08ABADA, 4 GB). A TSOP 48 socket was used to expose and connect the pins to the PMOD GPIO pins of a Digilent Zybo 7000 FPGA SoC device as shown in Fig. 3. The setup enabled high-speed data acquisition and analysis of memory degradation.

Results and Discussion:

A sample grayscale image was converted into a binary stream and was used to perform Program Erase Operations on the NAND memory. The image was broken into 10 segments and stored in 10 blocks and repeated on 10 copies of the image, totaling 100 blocks. Each block was subjected to 50,000 Program Erase cycles, close to the endurance limit of the NAND cells, to induce significant oxide damage. To induce bit failure, we use the Program Stress (PS) [1] operation to stress the cells by repeatedly programming the entire page with '1's. PE cycles were performed in increments of 10,000 on the blocks and after every 10,000 PE cycles the FBC was evaluated by subjecting the blocks to 5,000 Program Stress cycles in increments of 1000 cycles. The variation of FBC with respect to the number of Program Stress cycles are shown in Fig 4. and with respect to the number of Program Stress cycles in Fig. 5. The overall variation of FBC across different blocks with respect to the number of PE and PS cycles is shown in Fig. 6. This indicates that 2D NAND Flash devices exhibit progressively high Bit Failure due to Program Erase cycling followed Program Stressing.

References

[1] S. Sakib, A. Milenković and B. Ray, "Flash Watermark: An Anticounterfeiting Technique for NAND Flash Memories," in IEEE Transactions on Electron Devices, vol. 67, no. 10, pp. 4172-4177, Oct.2020

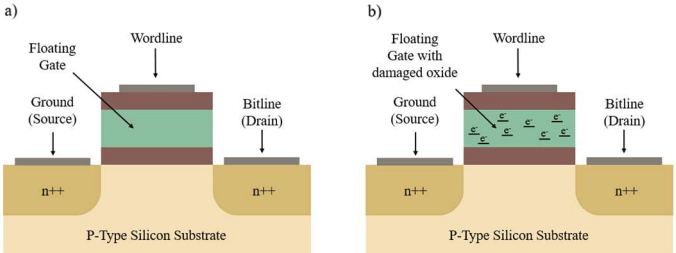

Fig. 1 Typical architecture of a NAND Flash cell. (a) represents a fresh cell and (b) represents a cell with oxide degradation due to trap generation from the repeated program erase cycles. The traps when filled by electrons increase the intrinsic V_t of the erased cells.

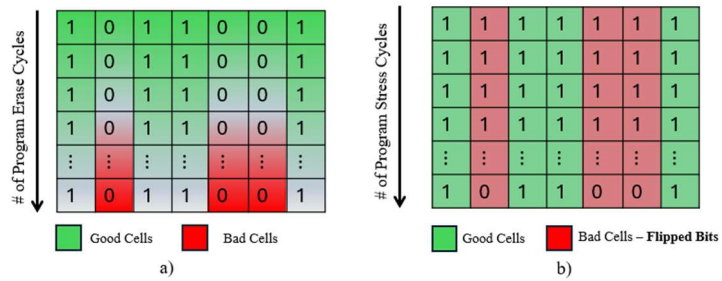

Fig. 2 (a) Generation of Bad Cells through repeated Program Erase operations due to damage to the oxide layer. (b) Inducing Bit Failure in damaged/bad cells through repeated Program Stress Operations. In a program stress operation, a page is repeatedly programmed with all '1's.

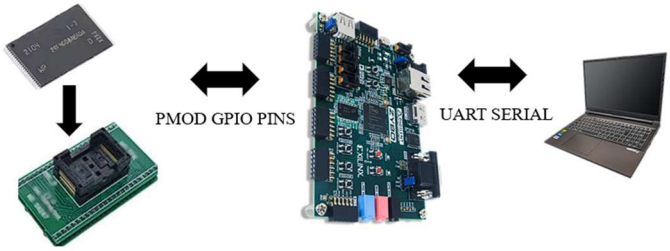

Fig. 3 Schematic of the measurement setup used to perform Program Erase and Program Stress operations on the NAND Flash Memory. The high-speed program and erase operations are controlled via Digilent Zybo 7000 SoC FPGA Board.

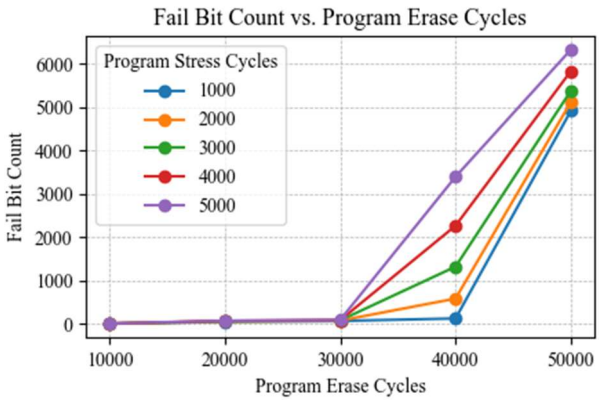

Fig 4. Fail Bit Count values with respect to the number of Program Erase cycles and Program Stress cycles. The blocks were cycled from 10,000 to 50,000 Program Erase Cycles in increments of 10,000 and

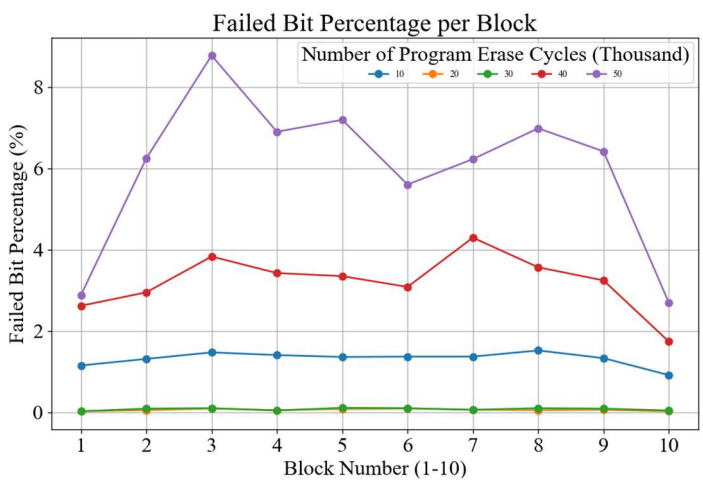

Fig. 5 Variation of Fail Bit Count percentage across different blocks with respect to the number of Program Erase cycles. All the blocks were subjected to 5000 Program Stress cycles and the Fail Bit Count was calculated from the data retrieved from the blocks. The initial and final blocks report a lower Fail Bit count due to the nature of the data chosen

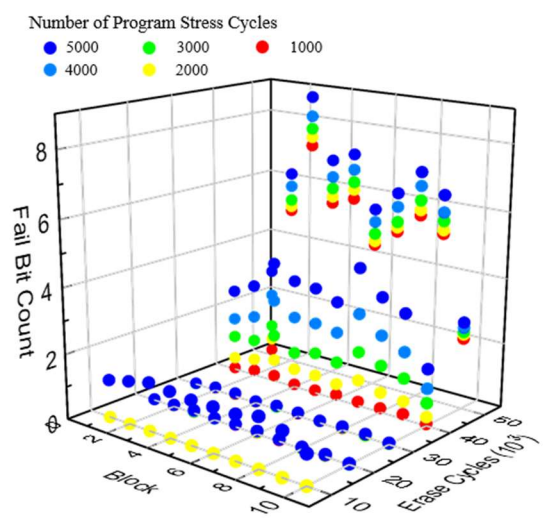

Fig. 6 Variation of Fail Bit Count percentage across different blocks with respect to the number of Program Erase cycles as well as Program Stress cycles. The highest FBC values are obtained at 50,000 PE and 5000 PS cycles across the different blocks.